

1 Summary

Model (Run-name): model1 (test)

Iteration: 11

TAC: 154557.47043323584

Took: 1.7203505039215088s

Total Time: 12.925496816635132s

Epsilons:

- ε_{LMTD} : 1e-06
- $\varepsilon_{Balancing}$: 1e-06
- ε_{Area} : 1e-06
- ε_{Beta} : 1e-06

Gurobi Tolerances:

- IntFeasTol: 1e-05
- FeasibilityTol: 1e-06
- OptimalityTol: 1e-06

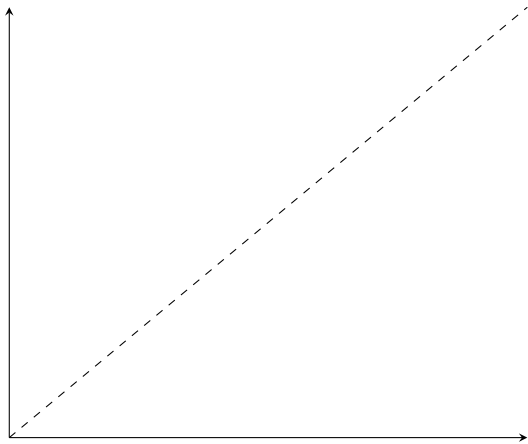
Gamsworld Solution: http://www.gamsworld.org/minlp/minlplib2/html/heatexch_gen1.html

2 Results

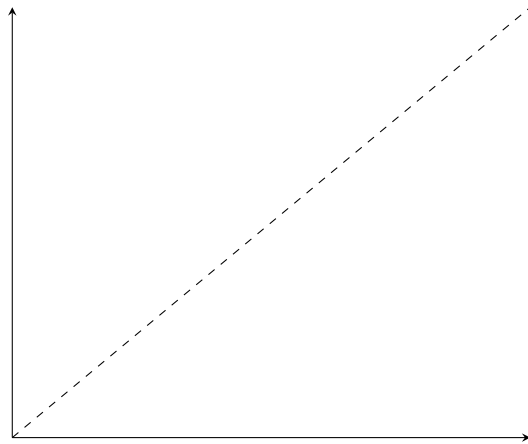
Variable	Value	Variable	Value	Variable	Value
$z_{1,1,1}$	1.0	$RecLMTD_{1,1,1}$	0.051752392959099974	$b_{1,1,1}^{H,out}$	918.439671309707
$z_{1,2,2}$	1.0	$RecLMTD_{1,2,2}$	0.01818995201843243	$b_{1,2,2}^{H,out}$	3854.964573062924
$z_{2,1,2}$	1.0	$RecLMTD_{2,1,2}$	0.027830858212904687	$b_{2,1,2}^{H,out}$	9392.553140405613
$z_{cu,1}$	1.0	$RecLMTD_{cu,1}$	0.0147651380494948	$b_{1,1,1}^{C,in}$	5001.956852711781
$z_{cu,2}$	1.0	$RecLMTD_{cu,2}$	0.009540062302996463	$b_{1,2,2}^{C,in}$	4550.0
$z_{hu,1}$	1.0	$RecLMTD_{hu,1}$	0.022449204186522416	$b_{2,1,2}^{C,in}$	6150.0
$A_{1,1,1}^\beta$	71.6637831223933	$q_{1,1,1}$	695.0354269370757	$b_{1,1,1}^{C,out}$	5696.992279648855
$A_{1,2,2}^\beta$	70.94081287188642	$q_{1,2,2}$	1950.0	$b_{1,2,2}^{C,out}$	6500.0
$A_{2,1,2}^\beta$	132.12228195956155	$q_{2,1,2}$	2407.446859594387	$b_{2,1,2}^{C,out}$	8557.446859594385
$A_{cu,1}^\beta$	4.269280092716329	$q_{cu,1}$	154.9645730629244		
$A_{cu,2}^\beta$	37.9454585315849	$q_{cu,2}$	1992.5531404056128		
$A_{hu,1}^\beta$	13.346967780023082	$q_{hu,1}$	497.5177134685367		
$A_{1,1,1}$	71.6637831223933	$t_{1,1}^{(H)}$	650.0		
$A_{1,2,2}$	70.94081287188642	$t_{1,2}^{(H)}$	580.4964573062924		
$A_{2,1,2}$	132.12228195956155	$t_{1,3}^{(H)}$	385.4964573062924		
$A_{cu,1}$	4.269280092716329	$t_{2,1}^{(H)}$	590.0		
$A_{cu,2}$	37.9454585315849	$t_{2,2}^{(H)}$	590.0		
$A_{hu,1}$	13.346967780023082	$t_{2,3}^{(H)}$	469.62765702028065		
$dt_{1,1,1}$	33.1678475645691	$t_{1,1}^{(C)}$	616.8321524354309		
$dt_{1,1,2}$	10.0	$t_{1,2}^{(C)}$	570.4964573062924		
$dt_{1,2,2}$	80.49645730629241	$t_{1,3}^{(C)}$	410.0		
$dt_{1,2,3}$	35.496457306292434	$t_{2,1}^{(C)}$	500.0		
$dt_{2,1,2}$	19.503542693707573	$t_{2,2}^{(C)}$	500.0		
$dt_{2,1,3}$	59.62765702028064	$t_{2,3}^{(C)}$	350.0		
$dt_{cu,1}$	65.49645730629244	$t_{1,1,1}^H$	370.0		
$dt_{cu,2}$	149.62765702028065	$t_{1,2,2}^H$	385.49645730629237		
$dt_{hu,1}$	63.167847564569115	$t_{2,1,2}^H$	469.62765702028065		
$f_{1,1,1}^H$	2.482269381918127	$t_{1,1,1}^C$	636.9614739994557		
$f_{1,2,2}^H$	10.0	$t_{1,2,2}^C$	500.0		
$f_{2,1,2}^H$	20.0	$t_{2,1,2}^C$	570.4964573062924		
$f_{1,1,1}^C$	8.764603507152087	$b_{1,1,1}^{H,in}$	1613.4750982467826		
$f_{1,2,2}^C$	13.0	$b_{1,2,2}^{H,in}$	5804.9645730629245		
$f_{2,1,2}^C$	15.0	$b_{2,1,2}^{H,in}$	11800.0		

3 *RecLMTD* Tangent Points

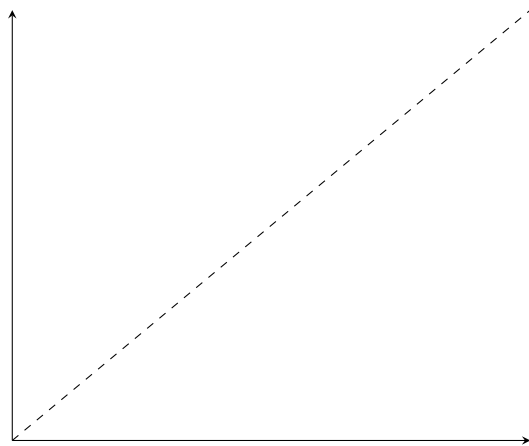
- $(1, 1, 1)$
 - Value: 0.051752
 - Error: 0.000000186277655
 - Relative error: 0.000003599388826
- $(1, 2, 2)$
 - Value: 0.018190
 - Error: 0.000005165259922
 - Relative error: 0.000283881650416
- $(2, 1, 2)$
 - Value: 0.027831
 - Error: 0.000020806639711
 - Relative error: 0.000747051920273
- $cu, 1$
 - Value: 0.014765
 - Error: 0.000000834501548
 - Relative error: 0.000056515176725
- $cu, 2$
 - Value: 0.009540
 - Error: 0.000000023490823
 - Relative error: 0.000002462328271
- $hu, 1$
 - Value: 0.022449
 - Error: 0.000000188934980
 - Relative error: 0.000008416039519



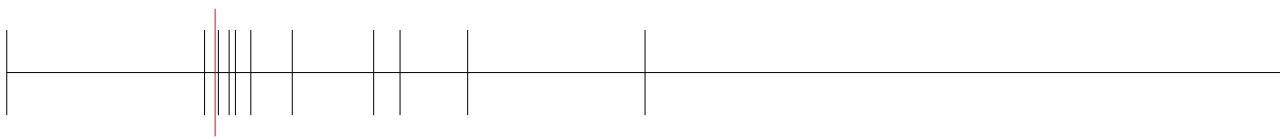
(a) *RecLMTD* - (1, 1, 1)



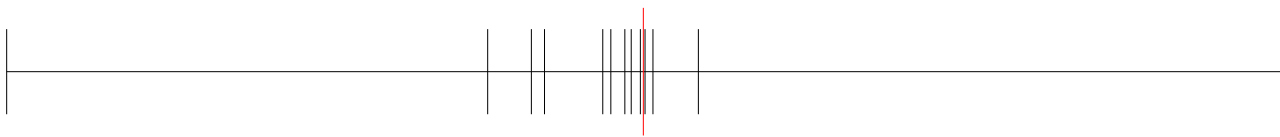
(b) *RecLMTD* - (1, 2, 2)



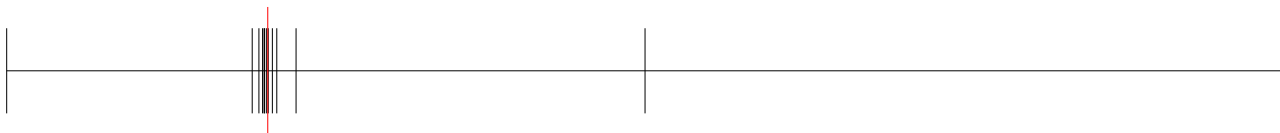
(c) *RecLMTD* - (2, 1, 2)



(a) *RecLMTD* - cu, 1



(b) *RecLMTD* - cu, 2



(a) *RecLMTD* - hu, 1

4 Balancing Breakpoints

4.1 $b^{H,\text{in}}$

- (1, 1, 1)
 - Value: 1613.4750982467826
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 5804.9645730629245
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 11800.0
 - Absolute error: 0.0
 - Relative error: 0.0

4.2 $b^{H,\text{out}}$

- (1, 1, 1)
 - Value: 918.439671309707
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 3854.964573062924
 - Absolute error: 4.547473508864641e-13
 - Relative error: 1.179640804131046e-16
- (2, 1, 2)
 - Value: 9392.553140405613
 - Absolute error: 0.0
 - Relative error: 0.0

4.3 $b^{C,\text{in}}$

- (1, 1, 1)
 - Value: 5001.956852711781
 - Absolute error: 1.7816021872095007
 - Relative error: 0.00035630794881091253
- (1, 2, 2)
 - Value: 4550.0
 - Absolute error: 0.0

- Relative error: 0.0
- (2, 1, 2)
 - Value: 6150.0
 - Absolute error: 0.0
 - Relative error: 0.0

4.4 $b^{C,out}$

- (1, 1, 1)
 - Value: 5696.992279648855
 - Absolute error: 114.27751071246257
 - Relative error: 0.02046988166909958
- (1, 2, 2)
 - Value: 6500.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 8557.446859594385
 - Absolute error: 1.8189894035458565e-12
 - Relative error: -2.1256216175113642e-16



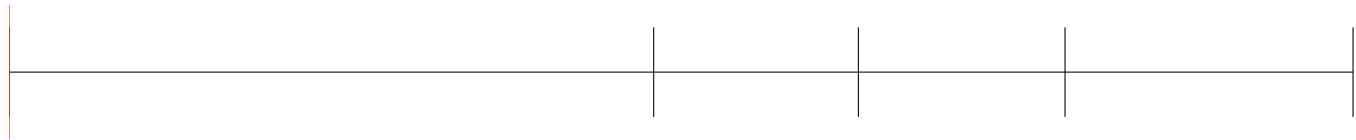
(a) $t^{(H)} - (1, 1)$



(a) $t^{(H)} - (1, 2)$



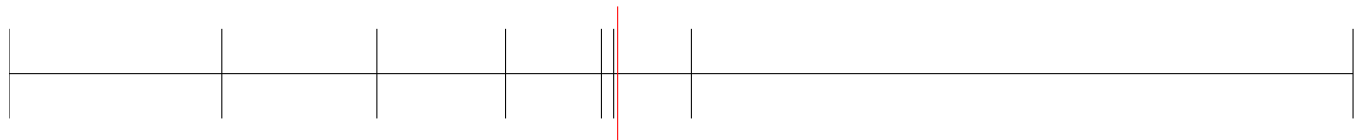
(a) $t^{(H)} - (2, 2)$



(a) $t^{(H)} - (1, 1, 1)$



(a) $t^{(H)} - (1, 2, 2)$



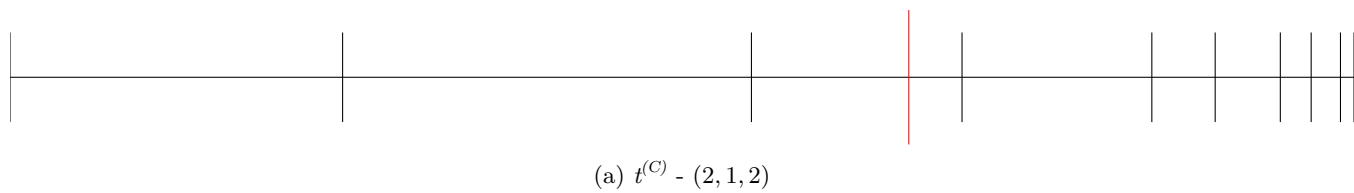
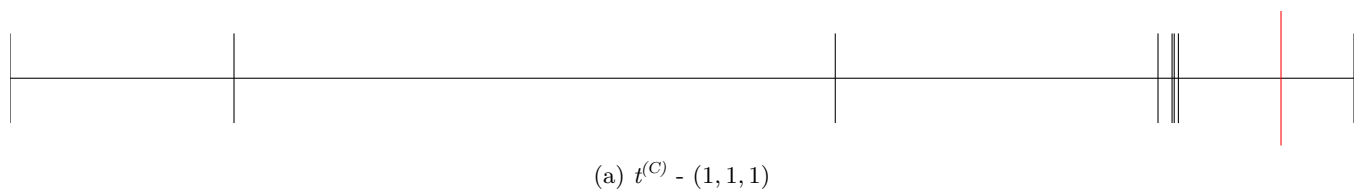
(a) $t^{(H)} - (2, 1, 2)$



(a) $t^{(C)} - (1, 2)$



(a) $t^{(C)} - (2, 3)$



5 Area Breakpoints

5.1 A

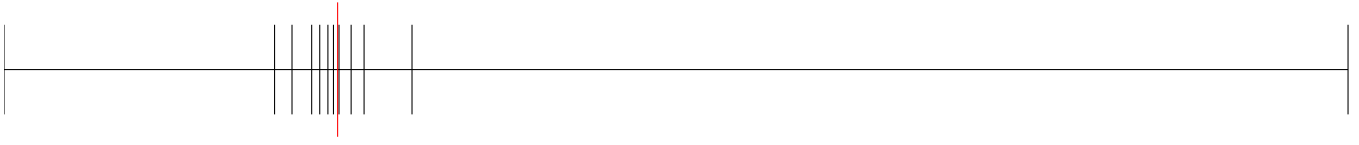
- $(1, 1, 1)$
 - Value: 71.663783
 - Error: 0.275709948293425
 - Relative error: 0.003832525592341
- $(1, 2, 2)$
 - Value: 70.940813
 - Error: 0.0000000000000057
 - Relative error: 0.000000000000001
- $(2, 1, 2)$
 - Value: 132.122282
 - Error: 1.880342449386546
 - Relative error: 0.014032131517426

5.2 A_{CU}

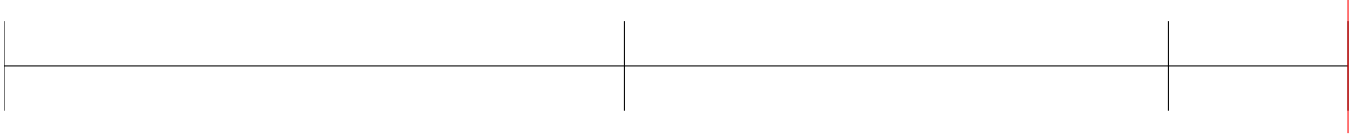
- $cu, 1$
 - Value: 4.269280
 - Error: 0.306866535393874
 - Relative error: 0.067057845897871
- $cu, 2$
 - Value: 37.945459
 - Error: 0.072703671416711
 - Relative error: 0.001912340502639

5.3 A_{HU}

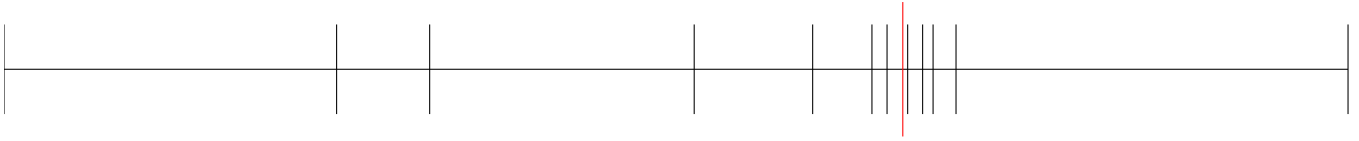
- $hu, 1$
 - Value: 13.346968
 - Error: 0.055684303257239
 - Relative error: 0.004154722730340



(a) $A^\beta - (1, 1, 1)$



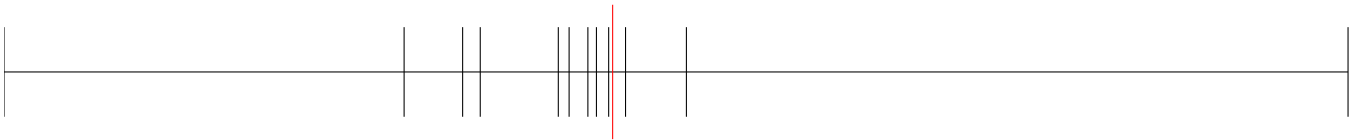
(a) $A^\beta - (1, 2, 2)$



(a) $A^\beta - (2, 1, 2)$



(a) $A^\beta - cu, 1$



(a) $A^\beta - cu, 2$

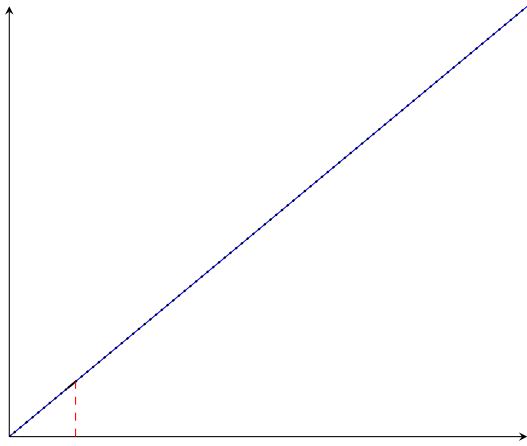


(a) $A^\beta - hu, 1$

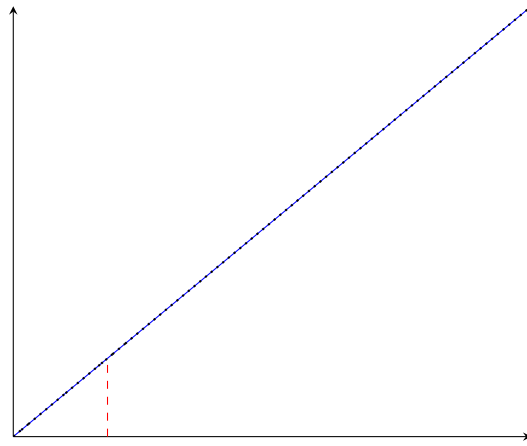
6 A^β Breakpoints

- (1, 1, 1)
 - q : 71.663783
 - $\hat{q}^\beta$: 71.663783
 - q^β : 71.663783
 - Breakpoints: [0, 5.0, 18.13353447520507, 60.38493803732795, 63.3468696286352, 65.40595072874316, 67.27362048222933, 69.14107665413653, 69.89902342903967, 70.30181758862284, 71.22933662920877, 560.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (1, 2, 2)
 - q : 70.940813
 - $\hat{q}^\beta$: 70.940813
 - q^β : 70.940813
 - Breakpoints: [0, 6.000977230027827, 11.262728503004022, 39.36524453694061, 64.01825678947372, 66.17325582523148, 69.34502840170047, 74.6496383881616, 84.1468297226002, 390.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (2, 1, 2)
 - q : 132.122282
 - $\hat{q}^\beta$: 132.122282
 - q^β : 132.122282
 - Breakpoints: [0, 9.898970403313276, 28.33333333333332, 42.69900440382105, 48.67452649027746, 79.83702775273511, 87.16107999267206, 118.88765652962957, 131.69760457629607, 147.48374761404892, 149.84689443310864, 720.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $cu, 1$
 - q : 4.269280
 - $\hat{q}^\beta$: 4.269280
 - q^β : 4.269280
 - Breakpoints: [0, 2.5450346292088537, 2.9819181007907165, 3.1809419553840232, 4.432280503220532, 5.0418124100090695, 7.702824927089129, 9.28003023700696, 13.56934656224007, 14.535545040173215, 181.61828057849593]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $cu, 2$
 - q : 37.945459
 - $\hat{q}^\beta$: 37.945459
 - q^β : 37.945459

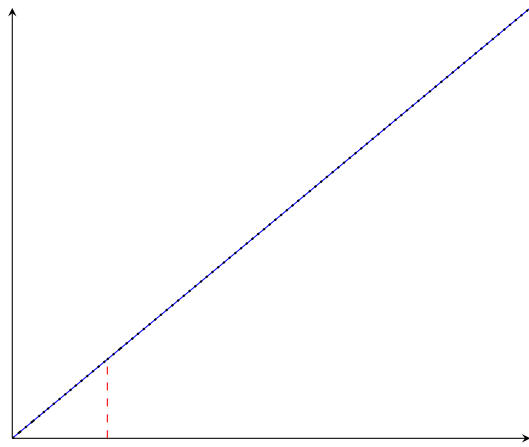
- Breakpoints: [0, 18.023935870161154, 21.04660840930274, 24.97364426355679, 30.189424467922265, 34.23465128608717, 36.670233878040335, 37.21616748032837, **37.47875108132638**, **38.21652191748154**, 38.88483572219945, 285.4001551947793]
- Error: 0.0000000000000000
- Relative error: 0.0000000000000000
- $hu, 1$
 - q : 13.346968
 - $\hat{q}^\beta$: 13.346968
 - q^β : 13.346968
 - Breakpoints: [0, 4.943755299006499, 12.544341108707222, 12.595096067947068, 12.917705220652834, 13.192252946941991, 13.22717920580809, 13.244985069124171, **13.30871858348944**, **13.468285806082731**, 14.09178614751222, 237.3002543523117]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000



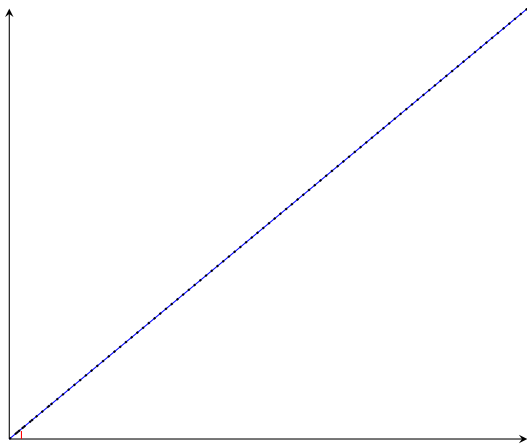
(a) $q^\beta - (1, 1, 1)$



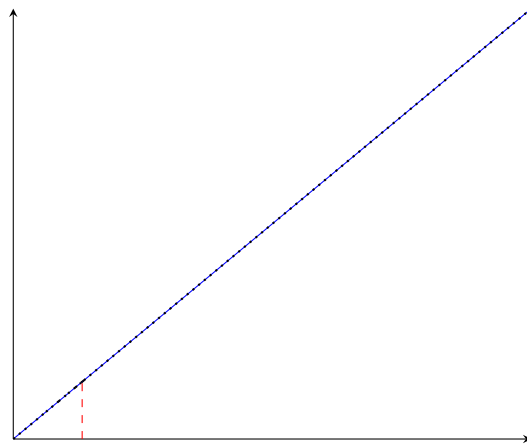
(b) $q^\beta - (1, 2, 2)$



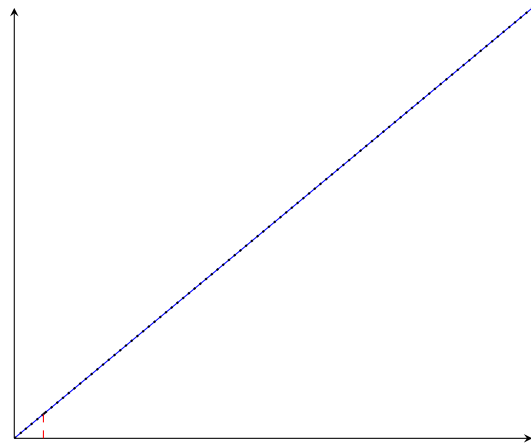
(c) $q^\beta - (2, 1, 2)$



(a) $q^\beta - cu, 1$



(b) $q^\beta - cu, 2$



7 Other variables

Variable	Value	Variable	Value	Variable	Value
$z_{1,1,2}$	-0.0	$f_{2,2,1}^C$	0.0	$b_{2,2,2}^{H,out}$	0.0
$z_{1,2,1}$	-0.0	$f_{2,2,2}^C$	0.0	$b_{1,1,2}^{C,in}$	0.0
$z_{2,1,1}$	0.0	$RecLMTD_{1,1,2}$	0.1	$b_{1,2,1}^{C,in}$	6500.0
$z_{2,2,1}$	0.0	$RecLMTD_{1,2,1}$	0.008106213167402865	$b_{2,1,1}^{C,in}$	3555.4900068826073
$z_{2,2,2}$	-0.0	$RecLMTD_{2,1,1}$	0.1	$b_{2,2,1}^{C,in}$	0.0
$z_{hu,2}$	-0.0	$RecLMTD_{2,2,1}$	0.00608	$b_{2,2,2}^{C,in}$	0.0
$A_{1,1,2}^\beta$	0.0	$RecLMTD_{2,2,2}$	0.004340277777777778	$b_{1,1,2}^{C,out}$	0.0
$A_{1,2,1}^\beta$	0.0	$RecLMTD_{hu,2}$	0.005549329826813255	$b_{1,2,1}^{C,out}$	6500.0
$A_{2,1,1}^\beta$	0.0	$q_{1,1,2}$	0.0	$b_{2,1,1}^{C,out}$	3555.4900068826073
$A_{2,2,1}^\beta$	0.0	$q_{1,2,1}$	0.0	$b_{2,2,1}^{C,out}$	0.0
$A_{2,2,2}^\beta$	0.0	$q_{2,1,1}$	0.0	$b_{2,2,2}^{C,out}$	0.0
$A_{hu,2}^\beta$	0.0	$q_{2,2,1}$	0.0		
$A_{1,1,2}$	0.0	$q_{2,2,2}$	0.0		
$A_{1,2,1}$	0.0	$q_{hu,2}$	0.0		
$A_{2,1,1}$	0.0	$t_{1,1,2}^H$	418.8196134269507		
$A_{2,2,1}$	0.0	$t_{1,2,1}^H$	650.0		
$A_{2,2,2}$	0.0	$t_{2,1,1}^H$	590.0		
$A_{hu,2}$	0.0	$t_{2,2,1}^H$	590.0		
$dt_{1,1,3}$	240.0	$t_{2,2,2}^H$	466.68393514135914		
$dt_{1,2,1}$	150.0	$t_{1,1,2}^C$	549.7862368345164		
$dt_{2,1,1}$	10.0	$t_{1,2,1}^C$	500.0		
$dt_{2,2,1}$	90.0	$t_{2,1,1}^C$	575.9306694020544		
$dt_{2,2,2}$	220.0	$t_{2,2,1}^C$	350.0		
$dt_{2,2,3}$	240.0	$t_{2,2,2}^C$	370.04544034995354		
$dt_{hu,2}$	180.0	$b_{1,1,2}^{H,in}$	0.0		
$f_{1,1,2}^H$	0.0	$b_{1,2,1}^{H,in}$	4886.524901753217		
$f_{1,2,1}^H$	7.517730618081873	$b_{2,1,1}^{H,in}$	11800.0		
$f_{2,1,1}^H$	20.0	$b_{2,2,1}^{H,in}$	0.0		
$f_{2,2,1}^H$	0.0	$b_{2,2,2}^{H,in}$	0.0		
$f_{2,2,2}^H$	0.0	$b_{1,1,2}^{H,out}$	0.0		
$f_{1,1,2}^C$	0.0	$b_{1,2,1}^{H,out}$	4886.524901753218		
$f_{1,2,1}^C$	13.0	$b_{2,1,1}^{H,out}$	11800.0		
$f_{2,1,1}^C$	6.235396492847913	$b_{2,2,1}^{H,out}$	0.0		